

Software-Hardware Co-Design for Energy-Efficient Continuous Health Monitoring via Task-Aware Compression

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Abstract—Low power consumption associated with data transmission and processing of wearable/implantable devices is crucial to ensure the usability of continuous health monitoring systems. In this paper, we propose a novel health monitoring framework where the signal acquired is compressed in a task-aware manner to preserve task-relevant information at the sensor end with a low computation cost. The resulting compressed signals can be transmitted with significantly lower bandwidth, analyzed directly without a dedicated reconstruction process, or reconstructed with high fidelity. Also, we propose a dedicated hardware architecture with sparse Booth encoding multiplication and the 1-D convolution pipeline for the task-aware compression and the analysis modules, respectively. Extensive experiments show that the proposed framework is accurate, with a seizure prediction accuracy of 89.70 % under a signal compression ratio of 1/16. The hardware architecture is implemented on an Alveo U250 FPGA board, achieving a power of 0.207 W at a clock frequency of 100 MHz.

Index Terms—Continuous health monitoring, energy-efficient, signal compression, on-device processing, deep neural networks.

I. INTRODUCTION

EARLY detection and prediction of diseases are critical to improving patient survival rate, however regular scheduled screening tests from hospital visits provide only a snapshot of the patient's health condition and fail to reveal instant information in a timely manner [1], [2]. For instance, the prediction of acute myocardial infarction [3] and epileptic seizure [4] could not be realized by counting on the result of the next scheduled screening test. Precision health, which aims to prevent diseases by early detection and follow-up treatment, relies on dynamic measurement of health parameters from health monitoring systems that

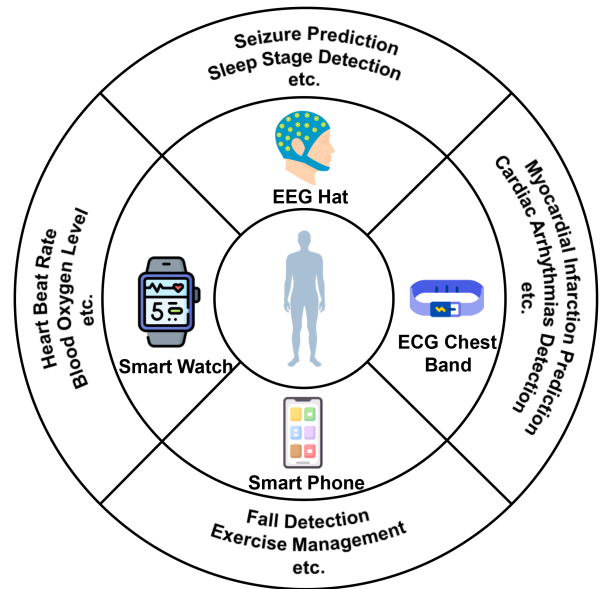


Fig. 1. Examples of popular health monitoring devices and corresponding applications.

are embedded into everyday life. Wearable and implantable devices are thus gaining more popularity among consumers. As seen from Fig. 1, a large variety of physiological signals such as blood pressure, heart rate, electrocardiogram (ECG), and electroencephalogram (EEG) are collected, transmitted or processed via daily wearable or implantable devices.

The advancement of deep neural network (DNN) based methodologies, which are capable of extracting discriminative features made possible accurate and complex physiological signal analysis in real-time, outperforming conventional approaches [4], [5], [6], [7]. However, power consumption induced by computation and transmission remains an enormous challenge for real-time health monitoring systems due to the miniature and battery constraints of monitoring devices [7]. In addition, the signal precision and transmission rate are limited by the transmission bandwidth. In-sensor signal compression [8] is a potential solution to reduce bandwidth requirements and resulting power consumption in the condition that the compression process can be implemented with low power and hardware cost. However, conventional compression algorithms such as least absolute shrinkage and selection operator (LASSO) [9], least

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angle regression (LARS) [10], and Sparse Bayesian Learning algorithms [11] rely on heavy computation and are hard to implement in resource-constrained devices.

Compressive sensing (CS) provides a relatively new technique that enables signal sampling and compression simultaneously at a sub-Nyquist sampling rate. The compression procedure could be defined as a matrix multiplication between a sensing matrix and the signal, which is hardware-friendly and computationally lightweight [12]. The research of CS resolves around two points: 1) How to design the sensing matrix for informative and efficient signal compression and 2) How to develop a reconstruction algorithm that best recovers the original signal from the corresponding measurements [13]. Current research endeavors are mainly focused on signal reconstruction neglecting the relationship between signal compression and downstream monitoring tasks. More importantly, existing CS approaches [14], [15] take the compression and reconstruction as two separate processes and compress signals in a task-agnostic fashion failing to capture informative features for various real-time health monitoring tasks. Consequently, compressed signals (randomized CS measurements) cannot be directly used for monitoring tasks analysis, and thus a complex reconstruction process is required before applying analysis to the signals. This, in turn, binds existing frameworks to a fixed paradigm where signals are compressed at the sensor end only and then transmitted to the cloud and reconstructed for further analysis.

To mitigate the aforementioned discussed limitations and to provide reliable and energy-efficient continuous health monitoring systems, this work presents a novel end-to-end framework for joint signal compressing, analysis, and reconstruction. Compared to traditional CS approaches, the compression technique proposed in our CHM monitoring framework is parametric and is directly applicable to unseen data without the need to solve an optimization problem on the new data. The sensing matrix learned by our proposed framework captures health monitoring task-relevant features and is optimal for both signal analysis and signal reconstruction for potential expert visualization and record keeping. The learned sensing matrix could be deployed on wearable or implantable devices as a generic plug-and-play solution to compress and transmit signals in a power-efficient manner. Moreover, the proposed framework is flexible where the monitoring task related analysis could act directly on the compressed signal on the end monitoring devices where transmission is unavailable.

We summarize the main contributions of this paper as follows:

- A novel deep learning based energy-efficient CHM framework which jointly solves signal compression, analysis, and reconstruction in a end-to-end manner.
- A task-aware compression module which reduces transmission and analysis energy consumption and enables task analysis directly on the compressed signal; A monitoring task analysis module that extracts features at various granularity levels; A reconstruction network with adaptive architecture based on different signal compression ratios.
- A FPGA implementation with sparse Booth encoding multiplication to perform efficient matrix multiplication and the 1-D convolution pipeline to maximize data reuse.

- We demonstrate the effectiveness of the proposed framework and the FPGA implementation results on the monitoring task of seizure prediction on a popular open-source dataset. Results show that the proposed framework outperforms state-of-the-art approaches under various compression ratios.

The remaining sections of this paper are organized as follows. In Section II, we first introduce previous research endeavors from compressive sensing and continuous health monitoring areas. We define and formalize our proposed framework in Section III. Section IV provides a dedicated hardware design. The effectiveness of our proposed framework is evaluated and validated in Section V. Lastly, the paper is concluded in Section VI.

II. RELATED WORK

A. Continuous Health Monitoring

Rapid advancement in low-power biomedical electronics and machine learning algorithms are revolutionizing the healthcare industry [16], [17]. Early detection and prevention of disease are considered promising solutions to promoting wellness. In particular, Continuous Health Monitoring (CHM), where wearable or implantable devices are utilized to capture, analyze or transmit various physiological signals, is envisioned as essential for proactive healthcare. Traditional health monitoring systems such as Holter [18] are only used for the long-term ECG recording [18] with signal processing and analysis performed offline. In recent years, researchers are stepping up the efforts to develop systems with real-time signal processing and analysis capabilities [19], [20]. Accelerometers are commonly used for physical activity detection associated fitness tracking [21]. Other vital signs such as oxygen saturation and blood glucose present a basic health condition portray of the user. ECG chest bands and EEG headsets enable some advance functionalities such as sudden heart failure prediction [22], seizure prediction [4] and emotion recognition [23].

However, the computation, storage, and transmission requirements for long-term CHM exceed the capabilities of existing devices. Power consumption for CHM systems can be roughly divided into signal acquisition, signal transmission, and signal analysis, where wireless signal transmission usually consumes the majority of the energy [24]. Battery recharging and replacement frequency are core factors for the usability of CHM systems. Consequently, performing on-sensor computation for signal processing and analysis without transmission or compressing the signal before transmitting could help reduce energy and storage consumption.

To this end, we propose a novel CHM framework that applies task-aware on-sensor signal compression, enabling energy-efficient signal transmission and analysis.

B. Compressive Sensing

Compressive Sensing (CS) is considered a breakthrough for signal compression following the famous Shannon sampling theorem. In CS, we desire to acquire a signal $x \in \mathbb{R}^n$ with only

m measurements where $m \ll n$. To ensure unique recovery, CS assumes that the underlying signal is approximately l -sparse in some certain basis, meaning that the l largest entries dominate others. The sparsity assumption holds naturally in many domains such as video processing [25], face recognition [26], [27], magnetic resonance imaging (MRI) acquisition [28], etc. In contrast to conventional sensing, CS adopts sensing matrices that are based on random functions rather than Dirac delta functions. Based on the available literature, the sensing matrix Φ can be either Bernoulli [29] or Gaussian [30]. The CS acquisition of CS can be mathematically expressed as:

$$z = \Phi x. \quad (1)$$

However, adopting random functions as sensing matrix fails to capture task-specific features from the original signal leading to deteriorated performance of downstream monitoring tasks analysis.

The CS reconstruction process from limited measurements is to solve the optimization task of an ill-posed inverse problem, which can be formulated as:

$$\arg \min_x \|\Phi x - z\|^2 + \mathfrak{R}(x), \quad (2)$$

where $\mathfrak{R}(\cdot)$ is a regularization prior term.

Various algorithms have been developed to solve the under-determined optimization problem for signal reconstruction. The Greedy algorithm [31] minimizes the least square error in each iteration to reconstruct the original signal in an iterative fashion. Matching pursuit [32] and orthogonal matching pursuits [33] are two popular algorithms due to their fast recovery speed. However, they are computationally costly when the signal is less sparse. Since the reconstruction objective is NP-hard, thus it is common to consider a convex relaxation via linear programming. Least angle regression (LARS), least absolute shrinkage and selection operator (LASSO), and basis pursuit (BP) [34] are some representative works.

To address the drawbacks of traditional CS methods, we propose to construct the sensing matrix, which captures task-relevant information during compression so that the compressed signal can be directly used for analysis and reconstruction.

III. PROPOSED CHM FRAMEWORK

We formalize and propose an end-to-end CHM framework. The principal objective of the proposed framework is to capture task-aware information from the original physiological signals and embed prior knowledge of the associated task into the sampled measurement during the signal compression process. The sampled measurement is then directly applicable to various monitoring tasks to reduce power consumption with minimum degradation in analysis performance for efficient real-time continuous health monitoring systems. Besides signal analysis, the proposed framework reconstructs the original signals from sampled measurements for subject case recording for further studying or expert visualization.

The proposed framework contains three modules, namely, a compression module $\mathcal{C}(\cdot)$, a reconstruction module $\mathcal{R}(\cdot)$ and an analysis module $\mathcal{A}(\cdot)$. Let $x \in \mathbb{R}^{N \times C}$ be physiological signals

of sequence length N and channel C , and y be its corresponding desired monitoring task-related output. A training phase is required to train all three modules for a specific monitoring task before the deployment phase. As illustrated in Fig. 2, during the training phase, the input signal x is first compressed by the compression module as follows:

$$z = \mathcal{C}(x), \quad (3)$$

where $z \in \mathbb{R}^{M \times C}$ represents the sampled measurement. The signal compression ratio r is then naturally defined as $\frac{M}{N}$ with $M < N$. The compressed signal z is then passed to the reconstruction module, and the analysis module to give the reconstructed signal $\hat{x}$ and the analysis result $\hat{y}$, respectively. During the actual deployment, the proposed framework also supports multiple deployment strategies based on different monitoring tasks and hardware requirements. One potential deployment setting could be that the compression module and the analysis module are deployed on the sensor. When a possible critical condition is detected, the sensor raises an alarm and then transmits the compressed signal to a cloud server for reconstruction and further analysis by experts with the assumption that transmission is more power-consuming than signal analysis. Another setting is that only the compression module is deployed on the sensor for signal compression purposes. The analysis and reconstruction modules are deployed on the cloud based on the compressed signal received to reduce power consumption further.

We implement all three modules with deep learning based neural networks. It is worth noticing that we aim to propose a generic framework and hence the structure of the neural networks can be of arbitrary designs. In this work, we provide a realization of these modules mainly with convolution-based networks. Next, we describe the architectures of the NNs in detail.

A. Compression Module

Conventional CS approaches adopt random Bernoulli or Gaussian matrices, while existing deep learning based CS methods mainly adopt a cascade of a few convolution layers as for signal compression. Huang et al. [35] adopted convolution layers with large stride without pooling layers stating that pooling operation in the compression stage causes severe information loss. Since the purpose of signal compression is to reduce the computation and transmission cost on wearable or implantable devices for continuous health monitoring systems, the compression process itself should be energy-efficient. Therefore, we choose trainable matrix $W \in \mathbb{R}^{N \times M}$ as the compression network. Then signal x^c is compressed to the sampled measurement y^c by applying matrix multiplication between the raw signal and the compression matrix, $y^c = Wx^c$. The compression matrix for each channel might vary if the underlying modalities are different across channels or the signals from each channel have a significant difference. Else, if the signal from different channels are expected to be similar and could be compressed to acquire the same features, the compressed

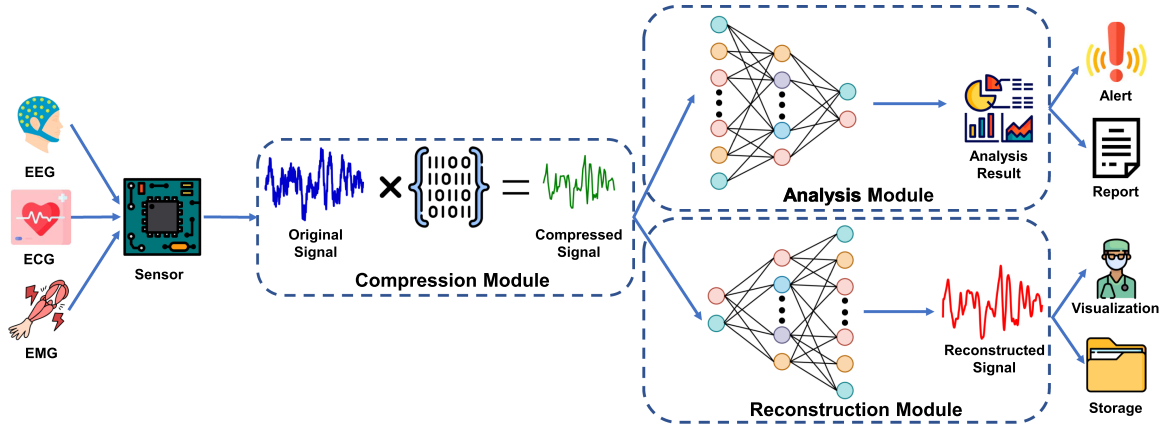


Fig. 2. A conceptual illustration of the training process of the proposed CHM framework. Physiological signals of various modalities are collected by wearable or implantable sensors and compressed via the compression module. The sampled measurements are then utilized for monitoring task analysis and signal reconstruction by the analysis module and reconstruction module, respectively. The compression matrix, analysis network and the reconstruction network are optimized simultaneously during training.

matrix could then be shared across channels to reduce memory consumption.

B. Analysis Module

Physiological signals contain essential information about human condition and behavioral activities and states and are the primary signal source for continuous health monitoring systems. To capture features of various granularity levels, the model implicitly extracts patterns of different frequencies. For instance, the alpha band of EEG signals is highly associated with eye closing and relaxation. Consequently, patterns of such frequency bands would be helpful for the sleep monitoring system. Inspired by the famous ResNet [36], we propose a strong baseline based on CNN with residual connections. 2-D convolution is the commonly adopted operation of most existing CNN-based approaches for physiological signal analysis. In other words, physiological signals from different channels are seen as images where the temporal dimension and channel dimension are considered as the height and width of an image. However, 2-D convolution has been widely used in the computer vision community due to the assumption that image pixels of a local region have high semantic correspondence and that convolution echoes this with different receptive field sizes. However, the aforementioned assumption does not hold for multichannel physiological signals. Signals from different channels might be collected from the same modality but from different electrodes or even from different modalities since the ordering of the channels, and the placement of electrodes are random such that adjacent channels are not guaranteed to be more functionally correlated than non-adjacent channels. 2-D convolution is therefore a poor choice due to its failure to obtain informative correlation among channels. In this work, we adopt 1-D convolution in lieu of 2-D convolution to maximally make use of the interactions among different channels by setting the kernel shape along the channel dimension to be the same as the input signal or intermediate features. Intuitively speaking, 1-D convolution takes information from all channels as input

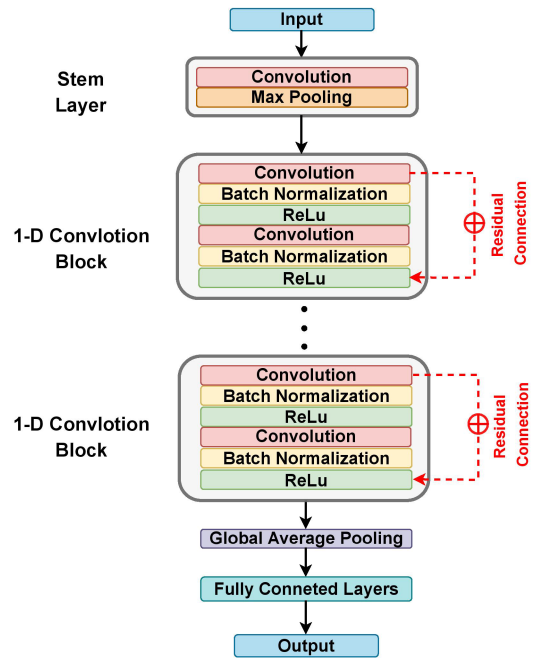


Fig. 3. Network architecture of the analysis module. The sampled measurements are first fed into the stem layer, which consists of a 1-D convolution layer and a max-pooling layer. The output intermediate features are further processed through 1-D convolution blocks. Each 1-D convolution block is composed of convolution, batch normalization, and activation operations. The dotted red line represents the residual connection. We use global average pooling and fully connected layers to aggregate information and generate task-specific analysis results.

and selects different channel correlation patterns with different kernels via network trainable parameters.

Next, we describe the proposed analysis architecture in detail. As shown in Fig. 3, the input raw signals are first fed to a stem layer consisting of a 1-D convolution layer and a max-pooling layer. The kernel size and the stride size of the convolution layer within the stem layer are set to relatively larger to reduce the computational complexity of the following

layers. The output from the stem layer is passed to a cascade of 1-D convolution blocks. Each 1-D convolution block contains 1-D convolution layers, with each followed by an activation layer. A batch normalization layer is further placed in between to avoid covariate shift and to ensure faster convergence. For the activation function, we choose the Rectified Linear Unit (ReLU). Define $conv_{l,k}$ to be the 1-D convolution layer of the l^{th} convolution block, we obtain the corresponding intermediate feature map:

$$f_{l,k} = \delta(BN(conv_{l,k}(f_{l,k-1}))), \quad (4)$$

where $f_{l,k}$ denotes the intermediate feature map, δ and BN represents the ReLU activation and batch normalization, respectively. We set $k = 2$ and $l = 4$ in this work. The output f_l of each convolution block is defined via residual connection:

$$f_l = f_{l,0} + \text{BottleNeck}(f_{l,2}), \quad (5)$$

where the *BottleNeck* operation reduces the channels of $f_{l,2}$ to be compatible with the number of channels of $f_{l,0}$.

The analysis result $\hat{y}$ is then provided as:

$$\hat{y} = \sigma(FC_2(\delta(FC_1(GAP(f_4))))), \quad (6)$$

where GAP stands for global average pooling and FC_i stands for the i^{th} fully connected layer. δ is the *LeakyReLU* function and σ is the *Softmax* function.

Finally, we define the loss function for the analysis task:

$$\mathcal{L}_{analysis} = \mathcal{L}(\hat{y}, y). \quad (7)$$

$\mathcal{L}_{task}$ denotes the objective function used to access the performance of the monitoring analysis task.

C. Reconstruction Module

In addition to monitoring analysis tasks, we also investigate signal reconstruction for additional conceivable applications which require the reconstruction of the original signal based on the sampled measurement. For instance, a clinician might rely on visualization of the original signal for diagnosis, or other analysis tasks could be performed based on computation of the original signal. Furthermore, we regard the task of the signal reconstruction as a regularizer to alleviate the overfitting problem of DL models due to limited dataset size. The reconstruction network is designed in an adaptive manner with flexible architecture composed of different numbers of up-sampling modules under various compression ratios to reduce reconstruction cost, as shown in Fig. 4. The up-sampling module is defined as:

$$\hat{f}_l = \delta(BN(conv_l(Up(\hat{f}_{l-1}))))). \quad (8)$$

Up-sampling module l takes input feature vector $\hat{f}_{l-1}$ as the input and outputs the intermediate reconstructed feature vector $\hat{f}_l$ where Up denotes linear interpolation. δ , BN and $conv_l$ denotes activation function, batch normalization layer and convolution layer respectively. Based on different lengths L of the original signal and a compression ratio r , a total number of $\lfloor \log_{\frac{1}{2}} r \rfloor + 1$ up-sampling modules are used in the reconstruction network. Each module doubles the input signal length, except that the last module maps the input signal to its original length.

Reconstruction Module with Compression Ratio r

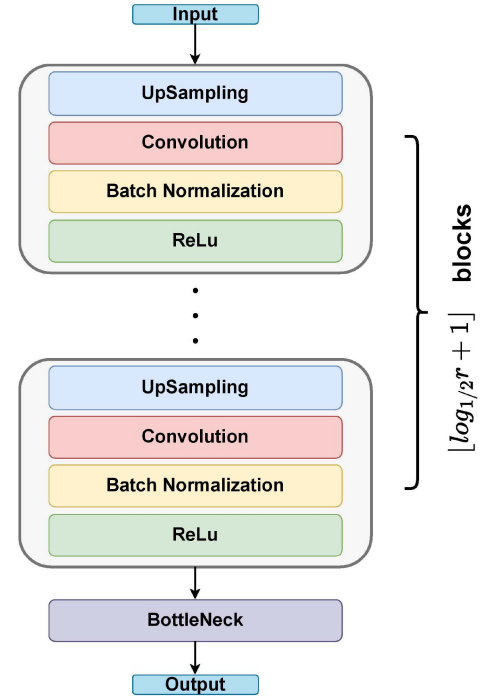


Fig. 4. Network architecture of the reconstruction module for signal compression ratio r . The reconstruction module is composed of $\lfloor \log_{\frac{1}{2}} r \rfloor + 1$ up-sampling modules in total. Each up-sampling module upsamples the input signal to twice the input signal length, with the exception that the last up-sampling module is used to map the length of the output signal to the original signal length. A *BottleNeck* operation is then adopted to map the number of channels of the reconstructed signal to that of the original signal.

To restore the signal to the original number of channels, we perform a *BottleNeck* operation on the signal output of the last up-sampling module. The reconstruction objective function is then defined as:

$$\mathcal{L}_{recon} = \mathcal{L}(\hat{x}, x). \quad (9)$$

Mean squared error loss is chosen as the objective function $\mathcal{L}_{recon}$ to access the quality of the reconstructed signal $\hat{x}$.

D. Joint-Objective Optimization

We simultaneously solve the task of compression, analysis, and reconstruction by optimizing the following joint objective:

$$\mathcal{L}_{joint} = \mathcal{L}_{task} + \lambda * \mathcal{L}_{recon}, \quad (10)$$

where λ is a scalar to balance the two objective functions. The sensing matrix is implicitly optimized during the joint-objective optimization process since both the analysis module and the reconstruction module take the output of the compression module as the input. The objectives of $\mathcal{L}_{task}$ and $\mathcal{L}_{recon}$ therefore ensures that the weights of the sensing matrix is optimized for both analysis and reconstruction via backpropagation. Consequently, the compressed signal is embedded with the prior knowledge of both the reconstruction and monitoring analysis tasks.

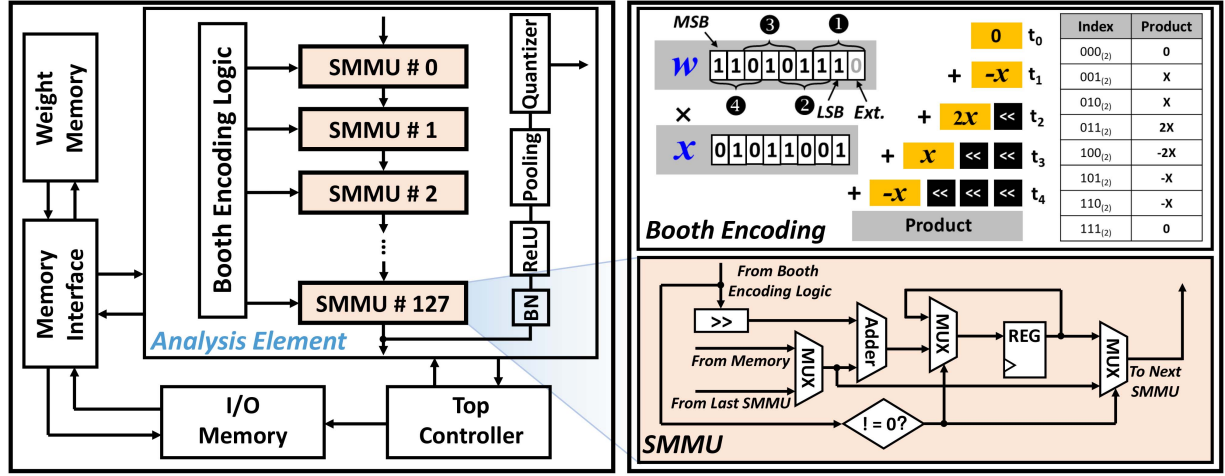


Fig. 5. Architecture of the proposed processor and the detailed architecture of the SMMU. BN: batch normalization module. ReLU: ReLU activation function module. Pooling: max pooling module. Adder: 24-bits adder.

IV. HARDWARE IMPLEMENTATION

With the deep learning-based compression sensing framework discussed in the previous session, we propose a dedicated hardware architecture to efficiently support the compression and analysis modules. We also verify the overall framework with FPGA implementation. The main goal of this hardware is to perform matrix multiplication and convolution operations used in compression and analysis modules, respectively. The proposed hardware utilizes Booth encoding multiplication with sparsity exploitation to perform efficient matrix multiplication and takes advantage of the proposed dedicated 1-D convolution pipeline to maximize data reuse for power reduction. The overall hardware architecture, detailed architecture of sparse matrix multiplication unit (SMMU), and workload mapping for different modules will be presented in this section.

A. Overall Hardware Architecture

The overall architecture of our proposed hardware is shown in Fig. 5. The data flow from memory to the analysis element (AE) via the memory interface, and the operations of the AE are controlled by the controller. Weight memory stores the weights, and I/O memory stores intermediate results and data just before or after external memory access. The memory interface is the bridge between the AE and memory, providing flexible memory reorder modes. The AE consists of 128 SMMUs and connects with the top controller for parameter configuration and interacts with the memory interface for memory access. Each SMMU can either work independently or connect with neighboring SMMUs to extend the accumulation pipeline. By changing the control signals, the accumulation pipeline can be adapted to the requirements of different matrix multiplications or convolution kernels.

B. Detailed Architecture of SMMU

The SMMU performs energy-efficient 8-bit matrix multiplication operations in compression and analysis modules using

the Booth encoding algorithm with sparsity exploitation. Fig. 5 depicts the Booth encoding multiplication used in 8-bits computations, which divides one 8-bit multiplier into four segments, and each segment is encoded by Booth encoding to obtain the corresponding value of the multiplicands. Then, one multiplication operation is replaced by several add operations and shift operations. Booth encoding logic encodes the multiplicand numbers by the four Booth multiplier numbers. The four encoded Booth multiplicand numbers are sent to 4 SMMUs, respectively. The multiplied numbers in each of the 4 SMMUs are shifted left the specified number of bits and then added with the partial results from the last SMMU. The multiplication results are accumulated together with the previously obtained results from the last four SMMUs.

There is native sparsity in matrix multiplication that a hardware can exploit. The sparsity rate is further enlarged when Booth multiplication is involved. This is due to the fact that fine-grained 3-bits encoding is utilized for encoding multiplicand numbers. In this work, we design the circuit for Booth-level sparsity exploitation. When each SMMU receives the encoded Booth multiplicand number, it first determines whether the number is zero. If the number is not zero, the 24-bit adder of the current SMMU will perform the addition operation, and the register will store the addition results. The addition result in the register that is calculated in the previous cycle will be passed to the next SMMU. If the receiving multiplicand number is zero, the adder of the current SMMU will perform the addition operation normally, but the register will be clock gated to remain the original value. At the same time, the partial results from memory or from the last SMMU will be passed directly to the next SMMU in place of the values in the register. By exploiting this Booth-level sparsity, the computational cycles and power consumption can be saved simultaneously.

When the result of the last SMMU is the partial result of the convolution operation, it will be sent to the next SMMU directly. However, if the result is the final result of the convolution operation, it will be sent to the corresponding operation modules based on the network architecture. The AE contains a

ReLU module for linear activation, a pooling module to support max pooling operation, and a quantizer module to quantize the results. The computation of the batch normalization layer is subsumed into the weights and biases of the convolution layer to avoid the computation of floating point numbers. The ReLU activation function is implemented by comparators combined with other corresponding combinational logic. For the Pooling module, we adopted additional pipeline registers combined with comparators to implement max pooling operation. The quantization module quantizes 24-bit intermediate results to 8-bit with configurable shifting module.

C. Workload Mapping

In order to map workloads to the hardware in both compression and analysis modules, the proposed hardware exploits two types of patterns. In this subsection, these two modes will be described in detail. Before performing calculations, the FPGA receives data from external DDR memory and stores in a buffer. The data need to be organized based on the computational paradigms of the two modules. Direct memory access (DMA) controls the data flow between the buffer and the memories of the proposed hardware. To efficiently exchange data, the Ping-Pong memory mode is applied for I/O memory.

1) *Compression Module*: The main operations in compression module is the multiplication of compression matrix $W \in \mathbb{R}^{N \times M}$ and raw signal matrix $x \in \mathbb{R}^{N \times C}$. This matrix multiplication is composed of multiple matrix-vector products. We split the two matrix multiplications into matrix W and multiple channel vector $x_i \in \mathbb{R}^{N \times 1}$ products. Since the dimension N of the matrix W and vector x_i is larger than 32 (the value of maximum computing capacity), it is not possible to calculate the final result with one loading. Therefore, it is feasible to load a part of the matrix W and vector x_i to obtain partial results and then accumulate the overall partial results. The x_i is stored in the register of Booth encoding logic as multipliers. While the data in matrix W^c is encoded by the generated Booth multipliers. By utilizing this mapping pattern, the raw signal vector x_i can be reused M times.

2) *Analysis Module*: For workload mapping of the analysis module, each four SMMUs can perform a 1×1 convolution operation. Take 1×8 convolution as an example, every 32 SMMUs are configured to be connected together for partial results accumulation. Every 32 SMMUs can complete the convolution operations of one kernel, and the whole hardware can perform the operation of four convolutional kernels simultaneously with one weight loading. The weights are stored in the register of Booth encoding logic as multipliers. While the input feature maps are encoded by the generated Booth multipliers. We design the pipeline specifically for 1-D convolution to maximize data reuse. Fig. 6 shows the data reuse in 1-D convolution pipeline. Due to the reusability of weights, the Booth multipliers are able to remain unchanged for different input feature maps until the computation of one kernel is completed. In addition, the input feature maps and partial sums are reused across SMMUs.

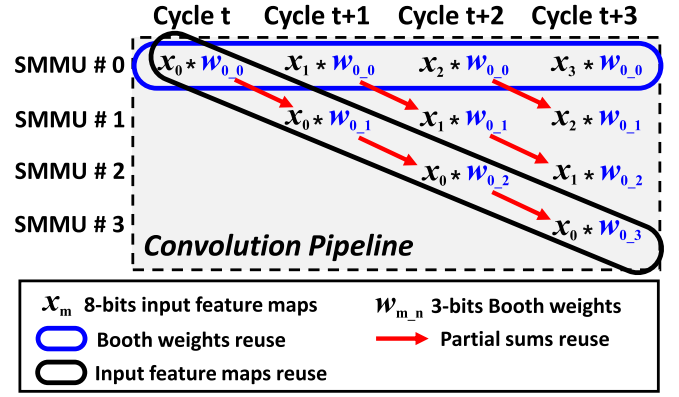


Fig. 6. Data reuse in 1-D convolution pipeline.

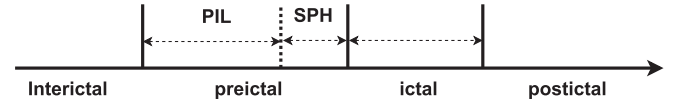


Fig. 7. Definition demonstration of seizure prediction horizon (SPH) and preictal interval length (PIL).

V. EXPERIMENTAL RESULTS

We showcase our proposed framework by designing a seizure prediction system to demonstrate effectiveness. We first provide preliminaries on epilepsy seizures, define a seizure prediction system, and introduce the dataset used.

A. Seizure Prediction System

As a widespread chronic brain disease, epilepsy afflicts around 70 million individuals, of which approximately up to one-third are drug-refractory [37], [38], [39]. Patients are highly dependent on family members and caregivers due to the safety uncertainty caused by seizure onset. Patients lose control over body parts experiencing unconsciousness and movement disorders when seizures occur. A seizure prediction system is expected to generate a warning to the patients before the seizure onset to provide time for emergent preparation under risky scenarios such as the operation of heavy machinery.

A seizure prediction system could be abstracted into a binary classification problem of recognizing whether the patient is about to have a seizure within an upcoming time frame. The effectiveness of a seizure prediction system is largely determined by the definition of preictal interval length (PIL), lead seizure, and the seizure prediction horizon (SPH). An illustration of PIL and SPH is given in Fig. 7. PIL defines the length of the preictal period, while SPH indicates the preparation time for patients between the warning and seizure onset. The length of SPH is a trade-off between feeling anxious caused by a long waiting time or not having enough time for proper preparation. Since multiple seizures tend to occur within a short period of time, thus lead seizures are of more research significance in clinical scenarios [40]. The lead seizure is defined as any seizure that is preceded by a seizure-free period. In this work, the length

TABLE I
SUMMARY OF SELECTED PATIENTS

Patient ID	No. of lead seizures / total seizure	Sample Count
chb01	3/7	354
chb05	2/5	353
chb06	6/10	685
chb07	2/3	238
chb08	3/5	355
chb09	3/4	357
chb10	6/7	547
chb14	4/8	456
chb18	3/6	247
chb19	3/3	238
chb22	3/3	301

of SPH and PIL are set to five and thirty minutes, respectively, while the seizure-free interval is set to four hours to define lead seizure. We collect preictal states only before lead seizures.

B. Dataset and Pre-Processing

EEG, as an important indicator for tracking brain activities, is the most commonly adopted physiological signal in a seizure prediction system. CHB-MIT sEEG [41] is a publicly accessible database that contains sEEG signals from 22 epileptic patients. The database contains 637 recordings recorded by 22 bi-polar electrodes placed based on the globally recognized 10-20 system from 17 females and five males. Some cases have electrodes added or removed during the EEG acquisition process. We remove these cases' data to avoid feature inconsistency caused by the variable electrode setting during acquisition. Patients with more than one lead seizure and at least one preictal period of one hour are chosen in this work. A sliding window is adopted to generate samples from continuous sEEG signals. To alleviate data imbalance, interictal data points are sampled using a non-overlapping sliding window of 20 seconds, and preictal data points are sampled with the same window length but with 25% overlapping between two consecutive windows. The sampled data points are standardized before feeding into the framework. A summary of patients which qualify the requirements are listed in Table I.

C. Experimental Setting

We first elaborate on the setting of three network structure related hyper-parameters, namely, the convolution filter number of the first up-sampling module $\#filters_{recon}$, the convolution filter number of the first 1-D convolution layer in the stem layer $\#filters_{stem}$ and the size of the linear layer of analysis network $size_{fc}$. For simplicity, the number of filters used in each up-sampling module doubles the filter number of the previous module. Similarly, the number of filters used in each convolution layer of l th 1-D convolution block is set to $2 * l * \#filters_{stem}$. Adam optimizer is adopted with an initial learning rate of $5e - 4$.

We train each model for maximum of 150 epochs with an early stopping criterion of no improvement for ten consecutive epochs. $\#filters_{stem}$, $\#filters_{recon}$, $size_{fc}$ and batch size are set to 32, 32, 100 and 64, respectively. The kernel size of the convolution layer in the stem layer is set to eight, and the kernel size of all other convolution layers is set to four in the analysis module. The stride is set to 1 for all convolution layers. The kernel and stride size of the for each convolution layer in the reconstruction module is set to three and two, respectively. All three modules are implemented with Pytorch framework and trained on NVIDIA 2080Ti GPU for acceleration.

D. Comparison With State-of-the-Art

The effectiveness of our proposed CHM framework, on the test case of seizure prediction, is validated by reporting seizure prediction accuracy, sensitivity, and false prediction rate (FPR). We pick two methods that use the original signal as input, together with two methods that use statistics as input for comparison. All four state-of-the-art methods are deep learning based and listed as follows:

- *End-to-End approach* [44]: the network structure of End-to-End approach is also based on CNN which only applies temporal convolution operation in early-stage and applies both temporal and spatial convolution (channel-wise convolution) in the late-stage.
- *EEGNet* [42]: EEGNet adopts a compact network architecture based on CNN containing various types of convolution operations such as pointwise convolution, temporal convolution and separable convolution.
- *STFT CNN* [45]: the STFT CNN takes the Short Time Fourier Transform of the EEG signals as input to a 3-layer CNN.
- *Light-weight approach* [43]: the light-weight approach is a CNN based network structure that takes the Pearson Correlation Coefficient (PCC) [47] which is seen as synchronization features across all channels as input.

For a fair comparison, all baseline methods are implemented using Pytorch, carefully following the setting described in the original papers. We train and evaluate each baseline method following the same settings in Section V-C using data pre-processed and split the same way as described in V-B. A random Gaussian matrix is adopted to compress signals for other baseline approaches. In order to demonstrate the robustness of the proposed framework, we evaluate and report the average value of the metrics over selected patients under various compression ratios: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{16}$. The comparison result is given in Fig. 8. We also provide numerical value comparisons with compression ratio 1/16 in Table II. As can be seen, the seizure prediction system based on our proposed methodology demonstrates consistent improvement over other baseline methods under various compression ratios, indicating the effectiveness and robustness of the proposed CHM framework. Furthermore, with compression ratios varying from $\frac{1}{2}$ to $\frac{1}{16}$, our method produces a minimum accuracy degradation. We notice more performance degradation on Xu et al. [44] and Lawhern et al. [42]'s methods under signal

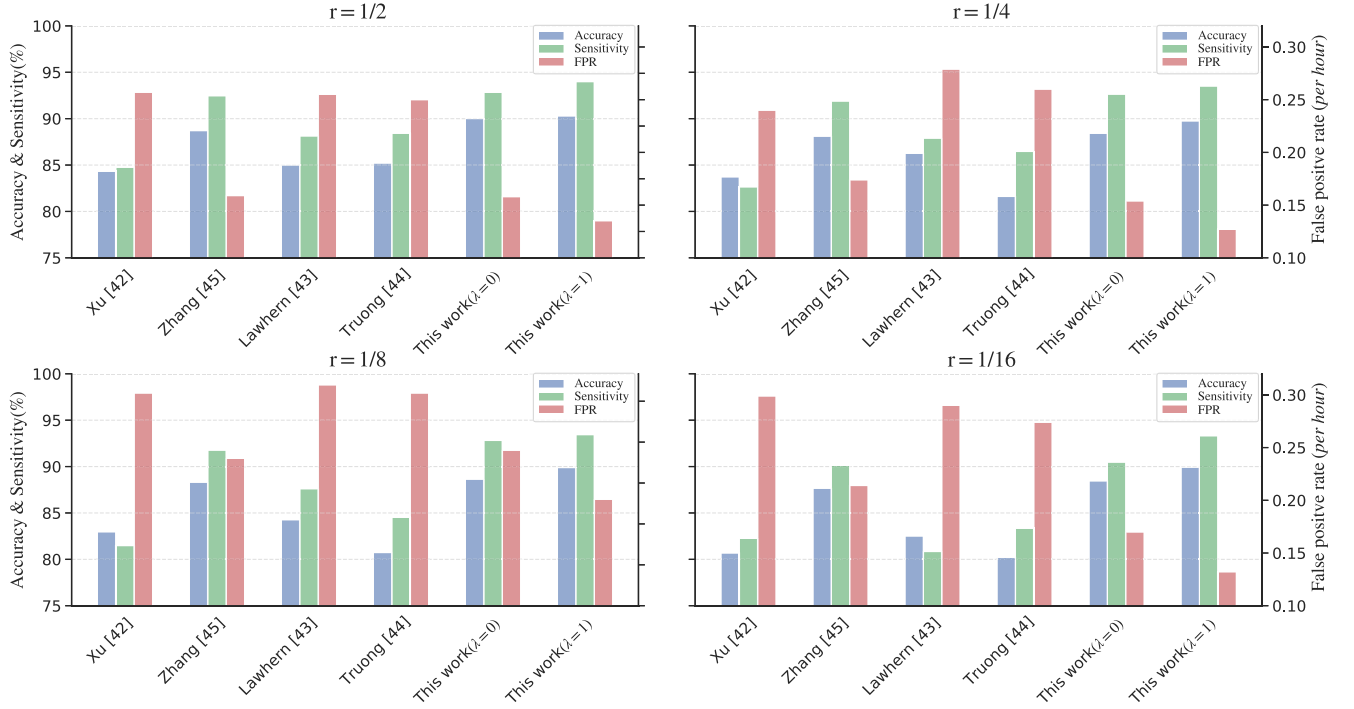


Fig. 8. Seizure prediction performance comparison between the system based on our proposed framework with state-of-the-art systems under various compression ratios. Higher accuracy, sensitivity and lower false positive rate indicate better performance.

TABLE II
SEIZURE PREDICTION PERFORMANCE COMPARISON UNDER THE COMPRESSION RATIO OF 1/16

	Accuracy (%)	Sensitivity (%)	FPR (/h)
Lawhern [42]	82.5	79.8	0.29
Zhang [43]	87.6	90.2	0.21
Xu [44]	80.6	82.2	0.30
Truon [45]	80.2	83.4	0.27
This work ($\lambda = 0$)	88.2	90.4	0.17
This work ($\lambda = 1$)	89.7	93.3	0.13

compression. These two methods take in raw signal as input and compression by multiplication with a random Gaussian matrix clearly captures no informative features for the underlying seizure prediction task. On the contrary, the other two methods by Truong et al. [45] and Zhang et al. [43] which use signal statistics as input show better stability under compression. However, manually extracting statistics from signals as input to the network discards potential task-relevant information and limits the feature extraction ability of neural networks. Our proposed framework jointly optimizes the sensing matrix for signal compression and task analysis so that the neural network extracts informative task-relevant features automatically. Moreover, we observe from Fig. 8 that setting $\lambda = 1$ (reconstruction is optimized together with the analysis network) yields better performance than setting $\lambda = 0$ (only the analysis network is optimized during training). This agrees with our hypothesis that the signal reconstruction task acts as regularization to the analysis task to alleviate possible over-fitting and lead to a more

generic learned representation. We evaluate the reconstruction performance with Peak Signal-to-noise Ratio (PSNR) and Pearson's Correlation Coefficient (PCC). Let x^c be the signal of channel c and $\hat{x}^c$ be its corresponding reconstructed version, PSNR is defined as:

$$PSNR(x, \hat{x}) = 10 \cdot \log_{10} \cdot \left(\frac{\max(x)}{MSE(x, \hat{x})} \right) \quad (11)$$

$$MSE(x, \hat{x}) = \frac{1}{C} \cdot \sum_{c=0}^C \|x^c - \hat{x}^c\|^2, \quad (12)$$

where C stands for the number of channels in total.

PCC is defined as:

$$PCC = \frac{1}{C} \cdot \sum_{c=0}^C \frac{(x^c - \mu^c)(\hat{x}^c - \hat{\mu}^c)}{\sqrt{(x^c - \mu^c)^2} \sqrt{(\hat{x}^c - \hat{\mu}^c)^2}}, \quad (13)$$

where μ^c and $\hat{\mu}^c$ denote the mean of the original signal and the reconstructed signal at channel c , respectively. The average PSNR and PCC values of all patients selected under various compression ratios are reported in Table IV. Besides quantitative results, a qualitative visual comparison of the reconstructed signal against the original signal is provided in Fig. 9. It is seen that the reconstructed signal recovers the visual characteristics of the original signal to a large degree.

Table III compares clock frequency, the monitoring task, performance, and FPGA resource utilization of our work with other state-of-art implementations. We demonstrate 92.9% sensitivity and 91.3% specificity in the more complicated task of seizure

TABLE III
FPGA IMPLEMENTATION RESULTS AND COMPARISON WITH PRIOR-ART PUBLICATIONS

	<i>This Work</i>	ASSCC'22 [20]	TBioCAS'21 [16]	GLSVLSI'18 [46]
FPGA Platform	Alveo U250	Alveo U250	Virtex-5	Zynq
Clock Frequency	100 MHz	75 MHz	86.73 MHz	100 MHz
Task	Seizure Prediction	Seizure Detection	Seizure Detection	Seizure Detection
Dataset	CHB-MIT	CHB-MIT	CHB-MIT	CHB-MIT
Algorithm	CNN	SNN	RDCSAE-IKRVFLN	CNN
Performance*	92.9%, 91.3%	91.09%	100%, 99.96%	86.98%
LUT	35631	21560	11963	—
FF	8672	20863	—	—
Power	0.207 W	0.787 W	1.59 W	2.07 W

*If only one metric is reported, the number stands for accuracy; If two metrics are reported, they represent sensitivity and specificity, respectively.

TABLE IV
RECONSTRUCTION PERFORMANCE UNDER DIFFERENT COMPRESSION RATIOS

Compression Ratio		$r = 1/2$	$r = 1/4$	$r = 1/8$	$r = 1/16$
Metrics	PSNR	40.34	36.91	36.30	33.81
	PCC	0.93	0.90	0.89	0.86

* PSNR = Peak Signal-to-noise Ratio; PCC = Pearson's Correlation Coefficient.

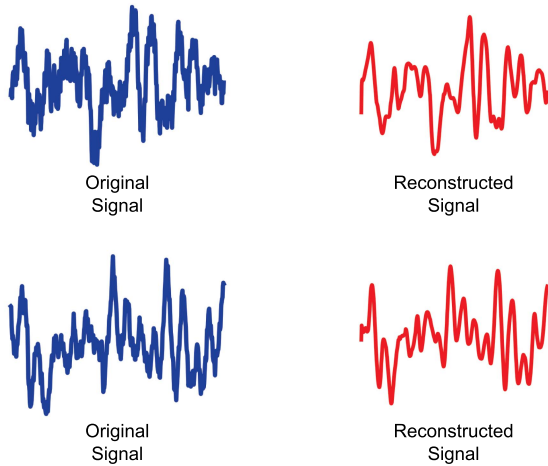


Fig. 9. Visualization comparison plotting of the reconstructed and the corresponding original signals.

prediction using the proposed CHM framework on the CHB-MIT EEG dataset. The maximum operating frequency of the proposed system is 100 MHz. We measured only 0.207 W overall power consumption and 0.05 W dynamic power of proposed framework on FPGA, and 1.2 W overall power consumption and 0.756 W dynamic power combined proposed framework with peripheral modules (DDR and XDMA). The proposed implementation trades off the use of register resources to achieve significant power savings compared to the SNN-based implementations [20]. It is worth noticing that [16] and [46] did

not provide complete information on hardware resources for instance FF.

VI. CONCLUSION

We presented an energy-efficient and reliable end-to-end health monitoring system with a joint signal compression, analysis, and reconstruction framework. The proposed framework compresses signals in a task-aware and power-efficient manner. The analysis and reconstruction modules are designed based on the DNN algorithm with a flexible architecture. The performance of our proposed framework is validated by a seizure prediction system. Results demonstrate that the proposed framework outperforms state-of-the-art works under various compression ratios. The hardware implementation on an Alveo U250 FPGA board shows that our proposed CHM framework can operate at a maximum clock frequency of 100 MHz with power consumption of 0.207 W, which is a significant improvement over the existing DNN-based works.

REFERENCES

- [1] R. Zhao, R. Yan, Z. Chen, K. Mao, P. Wang, and R. X. Gao, "Deep learning and its applications to machine health monitoring," *Mech. Syst. Signal Process.*, vol. 115, pp. 213–237, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0888327018303108>
- [2] S. S. Gambhir, T. J. Ge, O. Vermesh, R. Spitler, and G. E. Gold, "Continuous health monitoring: An opportunity for precision health," *Sci. Transl. Med.*, vol. 13, no. 597, 2021, Art. no. eae5383.
- [3] J. Kojuri, R. Boostani, P. Dehghani, F. Nowroozpour, and N. Saki, "Prediction of acute myocardial infarction with artificial neural networks in patients with nondiagnostic electrocardiogram," *J. Cardiovasc. Dis. Res.*, vol. 6, no. 2, 2015, Art. no. 51.
- [4] D. Wu, J. Yang, and M. Sawan, "Bridging the gap between patient-specific and patient-independent seizure prediction via knowledge distillation," *J. Neural Eng.*, vol. 19, 2022, Art. no. 036035.
- [5] D. Wu, S. Li, J. Yang, and M. Sawan, "neuro2vec: Masked Fourier spectrum prediction for neurophysiological representation learning," 2022, *arXiv:2204.12440*.
- [6] S. Zhao, J. Yang, and M. Sawan, "Energy-efficient neural network for epileptic seizure prediction," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 1, pp. 401–411, Jan. 2022.
- [7] S. Zhao, J. Yang, Y. Xu, and M. Sawan, "Binary single-dimensional convolutional neural network for seizure prediction," in *Proc. IEEE Int. Symp. Circuits Syst.*, 2020, pp. 1–5.

- [8] J. Yang and M. Sawan, "From seizure detection to smart and fully embedded seizure prediction engine: A review," *IEEE Trans. Biomed. Circuits Syst.*, vol. 14, no. 5, pp. 1008–1023, Oct. 2020.
- [9] S. Qaisar, R. M. Bilal, W. Iqbal, M. Naureen, and S. Lee, "Compressive sensing: From theory to applications, a survey," *J. Commun. Netw.*, vol. 15, no. 5, pp. 443–456, 2013.
- [10] M. Rudelson and R. Vershynin, "Sparse reconstruction by convex relaxation: Fourier and gaussian measurements," in *Proc. 40th Annu. Conf. Inf. Sci. Syst.*, 2006, pp. 207–212.
- [11] L. Wang, K. Lu, P. Liu, R. Ranjan, and L. Chen, "IK-SVD: Dictionary learning for spatial Big Data via incremental atom update," *Comput. Sci. Eng.*, vol. 16, no. 4, pp. 41–52, 2014.
- [12] H. Mamaghani, N. Khaled, D. Atienza, and P. Vanderghenst, "Compressed sensing for real-time energy-efficient ecg compression on wireless body sensor nodes," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 9, pp. 2456–2466, Sep. 2011.
- [13] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.
- [14] M. Shoaib, K. H. Lee, N. K. Jha, and N. Verma, "A 0.6–107 μ w energy-scalable processor for directly analyzing compressively-sensed EEG," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol. 61, no. 4, pp. 1105–1118, Apr. 2014.
- [15] M. Shooran, M. H. Kamal, C. Pollo, P. Vanderghenst, and A. Schmid, "Compact low-power cortical recording architecture for compressive multichannel data acquisition," *IEEE Trans. Biomed. Circuits Syst.*, vol. 8, no. 6, pp. 857–870, Dec. 2014.
- [16] M. Sahani, S. K. Rout, and P. K. Dash, "Epileptic seizure recognition using reduced deep convolutional stack autoencoder and improved kernel RVFLN from EEG signals," *IEEE Trans. Biomed. Circuits Syst.*, vol. 15, no. 3, pp. 595–605, Jun. 2021.
- [17] H. Elhosary, M. H. Zakhari, M. A. Elgammal, M. A. A. E. Ghany, K. N. Salama, and H. Mostafa, "Low-power hardware implementation of a support vector machine training and classification for neural seizure detection," *IEEE Trans. Biomed. Circuits Syst.*, vol. 13, no. 6, pp. 1324–1337, Dec. 2019.
- [18] N. J. Holter, "New method for heart studies," *Science*, vol. 134, no. 3486, pp. 1214–1220, 1961.
- [19] L. S. Vidyaratne and K. M. Iftikharuddin, "Real-time epileptic seizure detection using EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 11, pp. 2146–2156, Nov. 2017.
- [20] C. Fang, F. Tian, J. Yang, and M. Sawan, "A 217.8 MSOPs/W FPGA FPGA-based online learning SNN processor using unified event-driven structure and topology aware data reuse strategies," in *Proc. IEEE Asian Solid-State Circuits Conf.*, 2022, pp. 1–3.
- [21] A. Ignatov, "Real-time human activity recognition from accelerometer data using convolutional neural networks," *Appl. Soft Comput.*, vol. 62, pp. 915–922, 2018.
- [22] D. Lai, Y. Zhang, X. Zhang, Y. Su, and M. B. B. Heyat, "An automated strategy for early risk identification of sudden cardiac death by using machine learning approach on measurable arrhythmic risk markers," *IEEE Access*, vol. 7, pp. 94 701–94 716, 2019.
- [23] Y.-P. Lin et al., "EEG-based emotion recognition in music listening," *IEEE Trans. Biomed. Eng.*, vol. 57, no. 7, pp. 1798–1806, Jul. 2010.
- [24] A. M. Nia, M. Mozaffari-Kermani, S. Sur-Kolay, A. Raghunathan, and N. K. Jha, "Energy-efficient long-term continuous personal health monitoring," *IEEE Trans. Multi-Scale Comput. Syst.*, vol. 1, no. 2, pp. 85–98, Apr.–Jun. 2015.
- [25] S. Mun and J. E. Fowler, "Residual reconstruction for block-based compressed sensing of video," in *Proc. Data Compression Conf.*, 2011, pp. 183–192.
- [26] P. Nagesh and B. Li, "A compressive sensing approach for expression-invariant face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2009, pp. 1518–1525.
- [27] L. Qiao, S. Chen, and X. Tan, "Sparsity preserving projections with applications to face recognition," *Pattern Recognit.*, vol. 43, no. 1, pp. 331–341, 2010. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0031320309001964>
- [28] C. M. Sandino, J. Y. Cheng, F. Chen, M. Mardani, J. M. Pauly, and S. S. Vasanawala, "Compressed sensing: From research to clinical practice with deep neural networks: Shortening scan times for magnetic resonance imaging," *IEEE Signal Process. Mag.*, vol. 37, no. 1, pp. 117–127, Jan. 2020.
- [29] E. J. Candes and T. Tao, "Near-optimal signal recovery from random projections: Universal encoding strategies?," *IEEE Trans. Inf. Theory*, vol. 52, no. 12, pp. 5406–5425, Dec. 2006.
- [30] Z. Chen and J. J. Dongarra, "Condition numbers of Gaussian random matrices," *SIAM J. Matrix Anal. Appl.*, vol. 27, no. 3, pp. 603–620, 2005.
- [31] T. Blumensath and M. E. Davies, "Iterative hard thresholding for compressed sensing," *Appl. Comput. Harmon. Anal.*, vol. 27, no. 3, pp. 265–274, 2009. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1063520309000384>
- [32] S. Mallat and Z. Zhang, "Matching pursuits with time-frequency dictionaries," *IEEE Trans. Signal Process.*, vol. 41, no. 12, pp. 3397–3415, Dec. 1993.
- [33] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.
- [34] S. S. Chen, D. L. Donoho, and M. A. Saunders, "Atomic decomposition by basis pursuit," *SIAM Rev.*, vol. 43, no. 1, pp. 129–159, 2001.
- [35] J.-S. Huang, Y. Li, B.-Q. Chen, C. Lin, and B. Yao, "An intelligent eeg classification methodology based on sparse representation enhanced deep learning networks," *Front. Neurosci.*, vol. 14, 2020, Art. no. 808.
- [36] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [37] R. Fisher et al., "ILAE official report: A practical clinical definition of epilepsy," *Epilepsia*, vol. 55, pp. 475–482, 2014.
- [38] H. Zhang, D. Lai, C. Xie, H. Zhang, and W. Chen, "Directed-transfer-function based analysis for epileptic prediction," in *Proc. 9th Int. Congr. Image Signal Process. Biomed. Eng. Inf.*, 2016, pp. 1487–1491.
- [39] W. H. Organization, *Neurological Disorders: Public Health Challenges*. Geneva, Switzerland: World Health Organization, 2006.
- [40] H.-H. Chen and V. Cherkassky, "Performance metrics for online seizure prediction," *Neural Netw.*, vol. 128, pp. 22–32, 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0893608020301428>
- [41] A. Shoeb and J. Guttag, "Application of machine learning to epileptic seizure detection," in *Proc. 27th Int. Conf. Mach. Learn.*, 2010, pp. 975–982.
- [42] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for eeg-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, Jul. 2018, Art. no. 056013, doi: [10.1088/1741-2552/aac88c](https://doi.org/10.1088/1741-2552/aac88c).
- [43] Z. Shasha, D. Chen, R. Ranjan, K. Y. H. Tang, and A. Zomaya, "A lightweight solution to epileptic seizure prediction based on eeg synchronization measurement," *J. Supercomputing*, vol. 77, pp. 1–19, 04 2021.
- [44] Y. Xu, J. Yang, S. Zhao, H. Wu, and M. Sawan, "An end-to-end deep learning approach for epileptic seizure prediction," in *Proc. 2nd IEEE Int. Conf. Artif. Intell. Circuits Syst.*, 2020, pp. 266–270.
- [45] N. Truong et al., "Convolutional neural networks for seizure prediction using intracranial and scalp electroencephalogram," *Neural Netw.*, vol. 105, pp. 104–111, 2018.
- [46] C. Shea, A. Page, and T. Mohsenin, "SCALENet: A scalable low power accelerator for real-time embedded deep neural networks," in *Proc. Great Lakes Symp. VLSI*, 2018, pp. 129–134.
- [47] P. Schober, C. Boer, and L. A. Schwarte, "Correlation coefficients: Appropriate use and interpretation," *Anesth. Analg.*, vol. 126, no. 5, pp. 1763–1768, 2018.



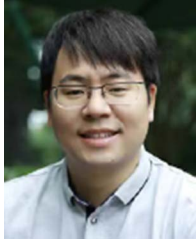
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